Predicting Next Day Gold Price

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Business goal and motivation

- **<u>Goal: Forecast the next trading day's gold price</u>**
- Trading firms use such models to execute high-frequency
- Risk management: by anticipating adverse price moves
- Macro indicator

Evaluation metric : MAPE

- **Mean Absolute Percentage Error** (MAPE)
- A - Actual
- F - Forecasted
- N - Sample size

$$MAPE = \Sigma((A-F)/A)*100/N$$

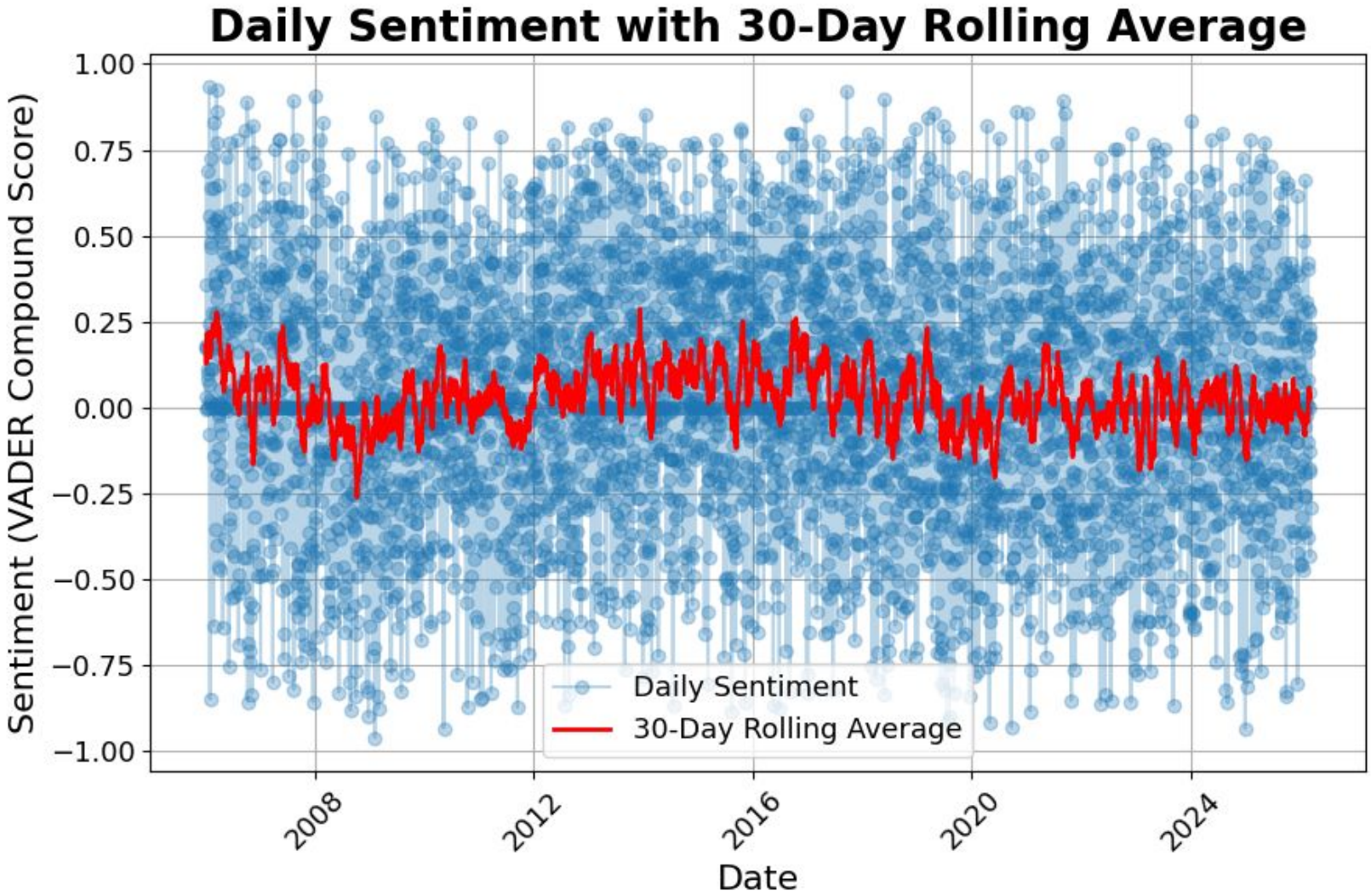
- MAPE expresses error as a percentage, making it independent of the absolute price level
- Aligns with how traders think

Data collection

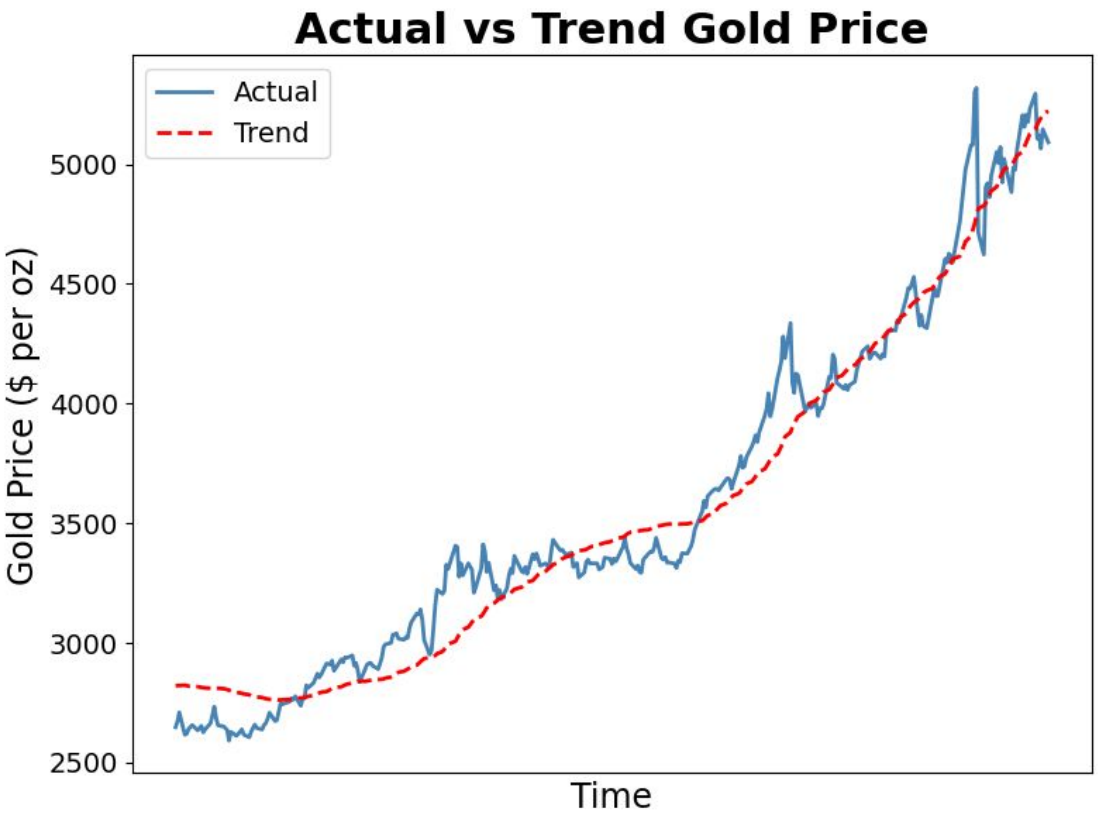
- **Sources:** Yahoo Finance, Federal Reserve Economic Data, New York Times
- Daily data for last 20 years (trading days)

- Gold Price
- US Dollar Index
- Crude Oil Prices
- S&P 500 volatility (Fear)
- S&P 500 index
- News sentiment
- Inflation Adjusted Return
- Expected Inflation
- Federal Funds Rate
- Consumer Price Index
- Money Supply

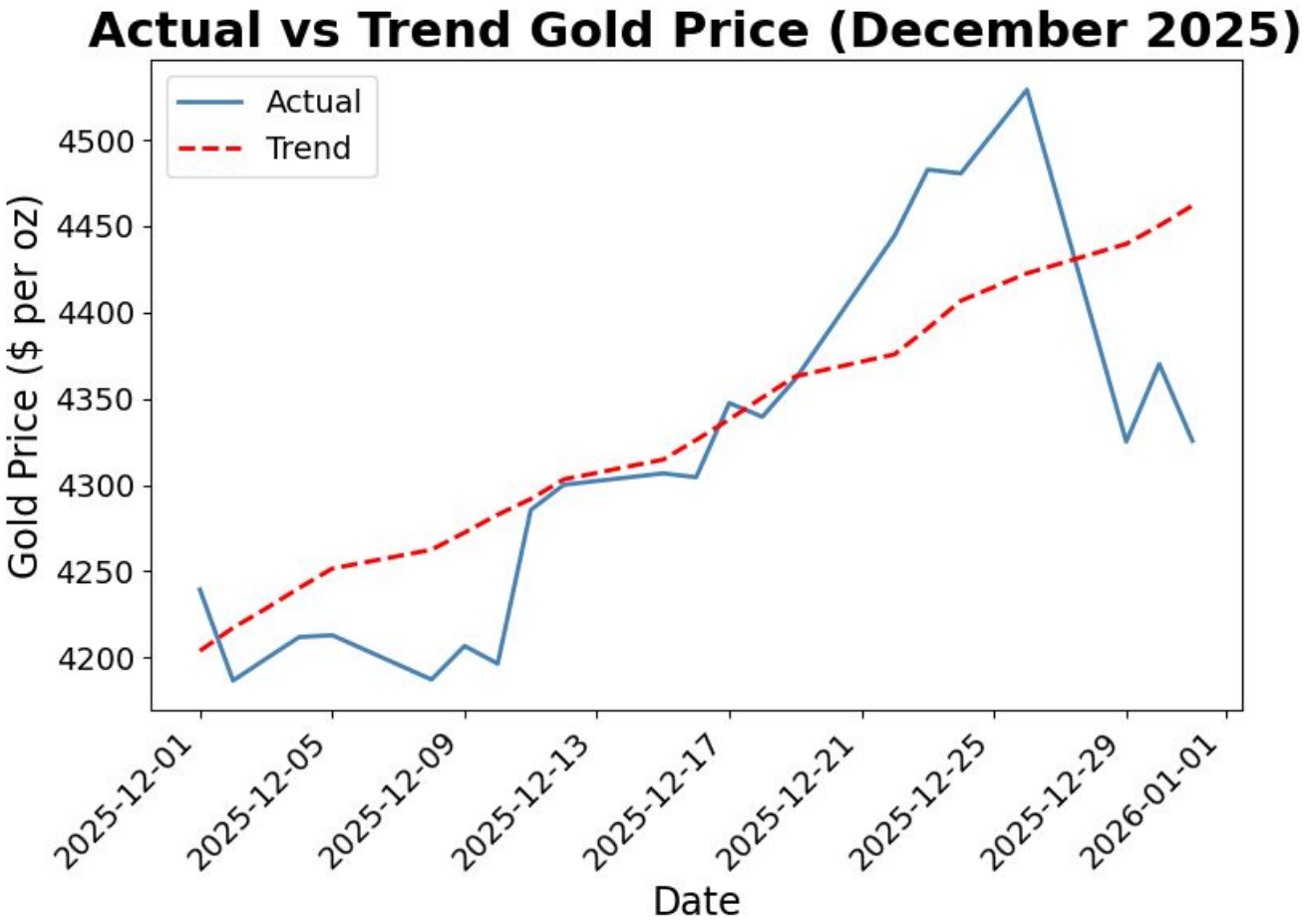
Gold Price Over Time
Gold Price ($ per oz)
5000
4000
3000
2000
1000
2008
2012
2016
2020
2024
Date


Daily Sentiment with 30-Day Rolling Average
Sentiment (VADER Compound Score)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Daily Sentiment
30-Day Rolling Average
2008
2012
2016
2020
2024
Date

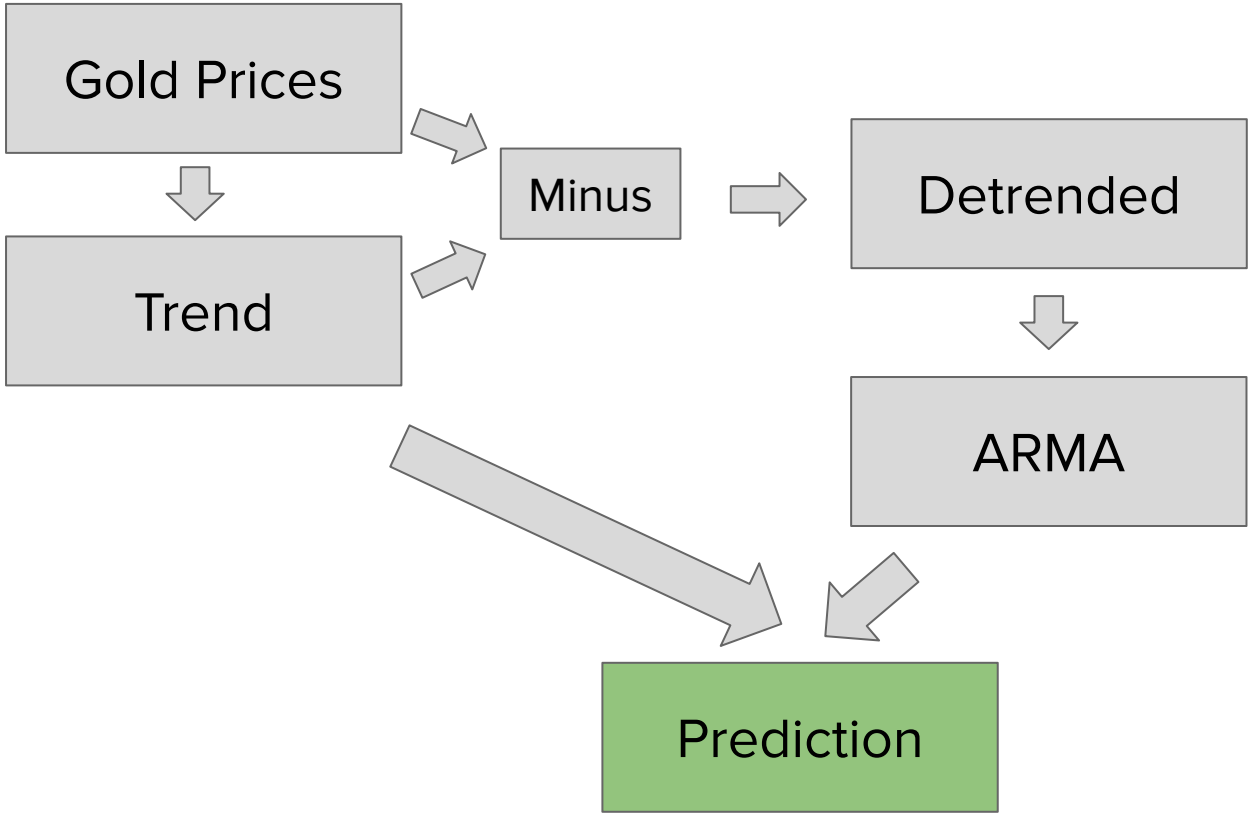
Baseline Trend Model: Linear Regression with Polynomial Features



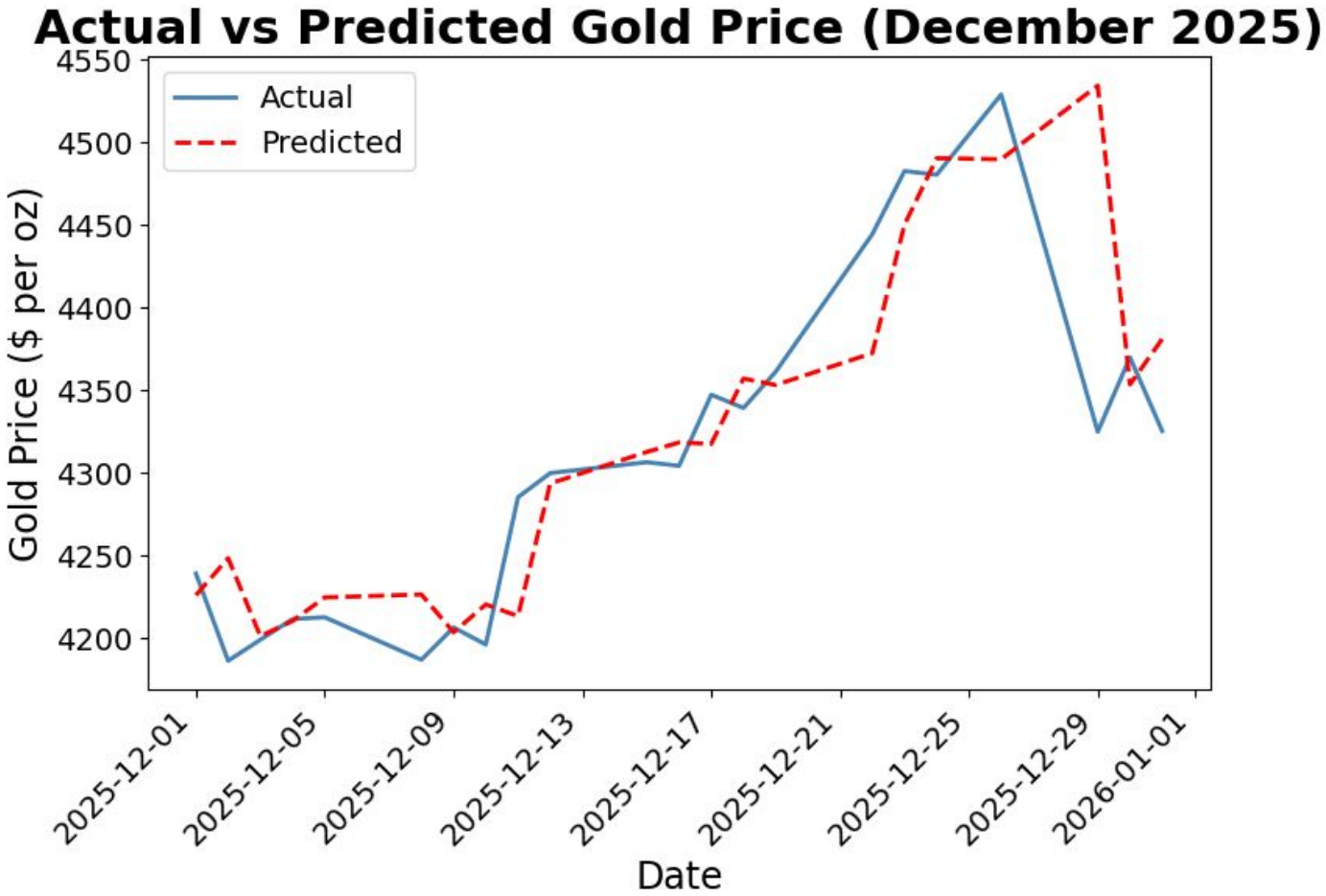
Baseline Trend Model | MAPE: 3.079



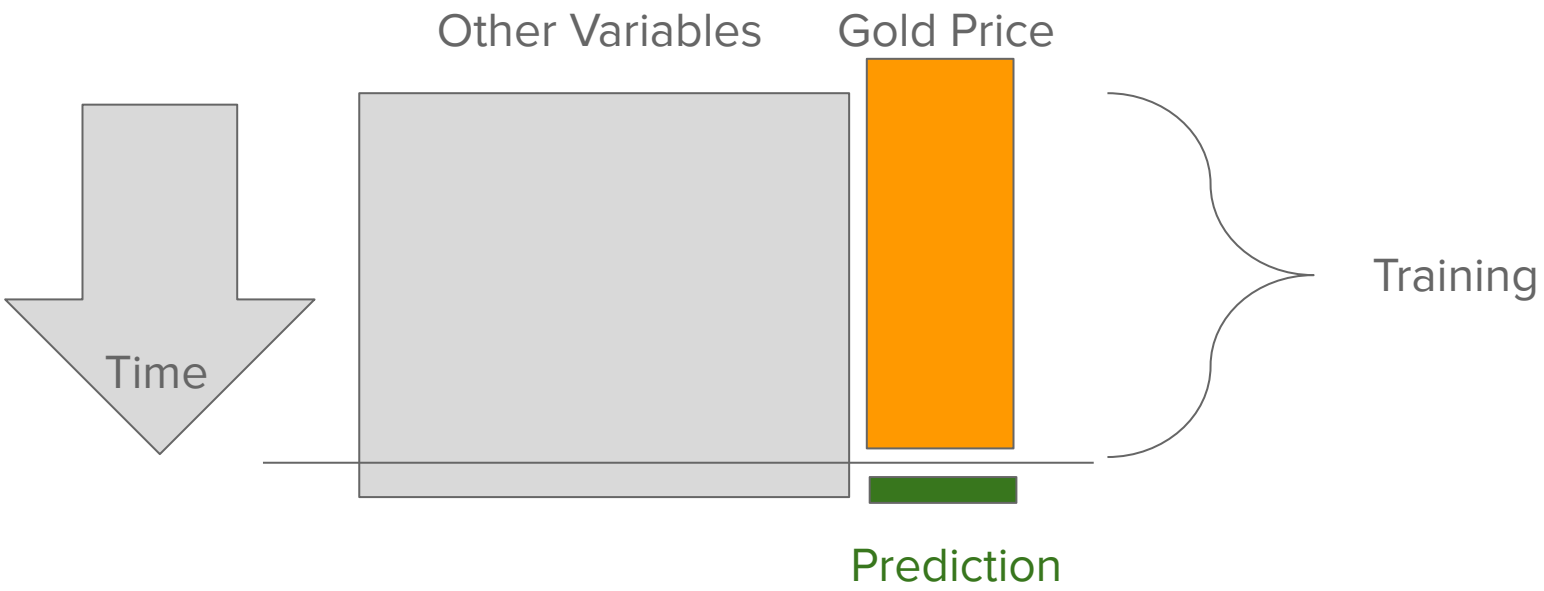
Improved Model: Trend + ARMA



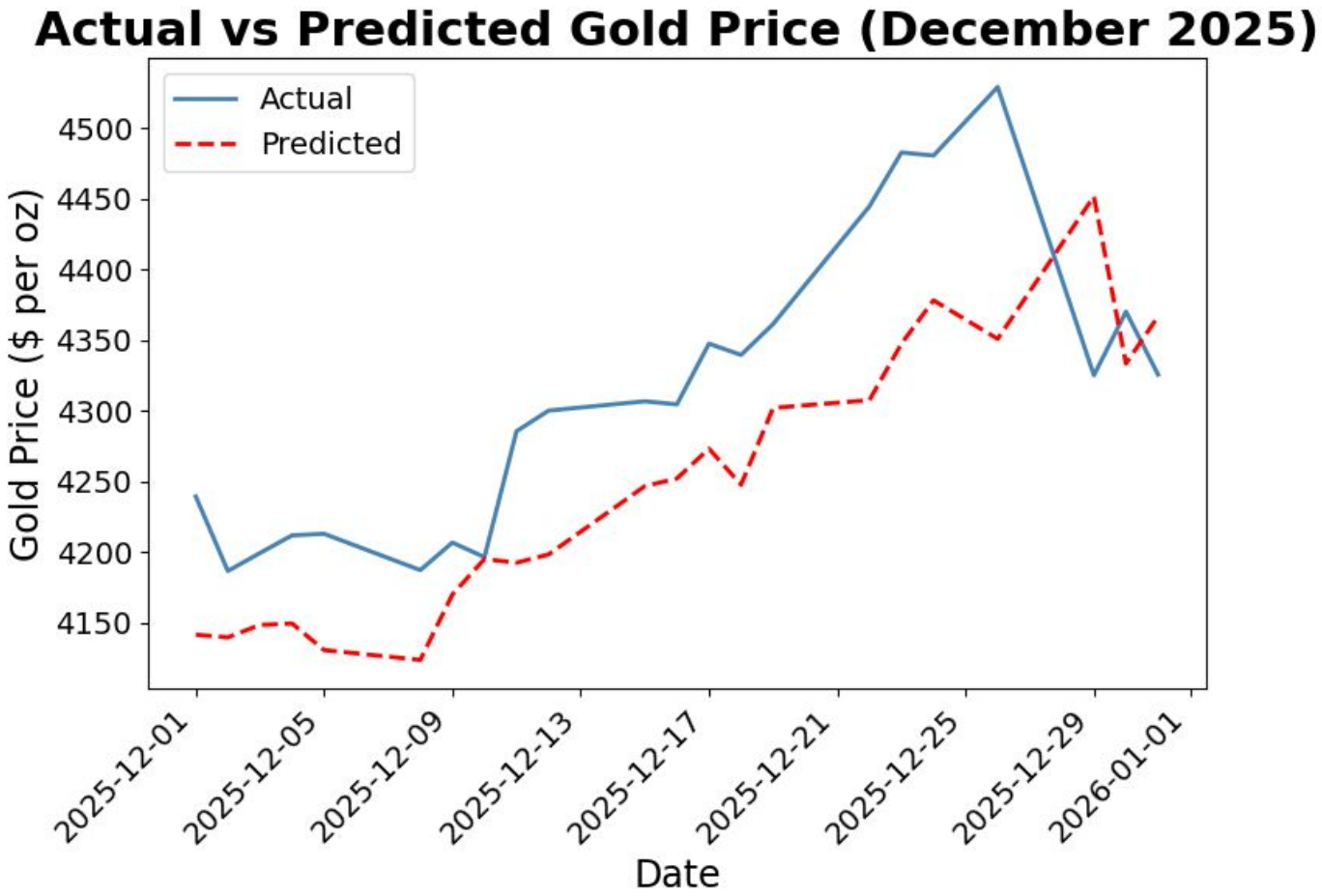
Trend + ARMA | MAPE: 0.967



Advanced Model: XGBoost



XGBoost Model | MAPE: 1.352



Models comparison

Model	MAPE (for 2025)
Baseline (trend)	3.079
Trend + ARMA	0.967
XGBoost with news sentiment	1.356
XGBoost without news sentiment	1.352

Web App: CI/CD Pipeline Implemented

Automatic continuous deployment every day with new incoming data

Future work

- Improve XGBoost code - Hyperparameter tuning
- Enhance news sentiment analysis - make it more specific
- Try out more advanced models
- App can be made more interactive
- Challenging due to too many external factors

Tech Stack

- jupyterlab
- fredapi
- yfinance
- matplotlib
- pandas
- numpy
- seaborn
- Scikit-learn
- vaderSentiment
- pmdarima
- xgboost
- streamlit

Thank you

My Linkedin: